Hölder Spaces

Let $U \subset \mathbb{R}^n$ be an open set.

<u>Def:</u> Let $C(\bar{U})$ denote the linear space of all continuous and bounded functions $u : U \to \mathbb{R}$. For $u \in C(\bar{U})$, define

$$\|u\|_{C(\bar{U})} = \sup_{x \in U} |u(x)|.$$

Thm: $\|\cdot\|$ is a norm on $C(\bar{U})$.

Pf: 1). Clearly, $\|u\|_{C(\bar{U})} \geq 0$ and equality holds iff

$u(x) = 0$ for all $x \in U$, i.e., $u = 0$.

2). For $u \in C(\bar{U})$ and $\lambda \in \mathbb{R}$,

$|\lambda||u(x)|$

$$\|\lambda u\|_{C(\bar{U})} = \sup_{x \in U} |\lambda u(x)| = |\lambda| \left(\sup_{x \in U} |u(x)| \right) = |\lambda| \|u\|_{C(\bar{U})}.$$

3). For $u, v \in C(\bar{U})$ and $x \in U$,

$$\underline{|u(x)+v(x)| \leq} \; |u(x)| + |v(x)|$$

$$\leq \sup_{y \in U} |u(y)| + \sup_{y \in U} |v(y)|$$

$$= \underbrace{\|u\|_{C(\bar{U})} + \|v\|_{C(\bar{U})}}_{\text{constant}}.$$

Taking the sup over all $x \in U$ gives

$$\sup_{x \in U} |u(x) + v(x)| \leq \|u\|_{C(\bar{U})} + \|v\|_{C(\bar{U})},$$

i.e., $\|u+v\|_{C(\bar{U})} \leq \|u\|_{C(\bar{U})} + \|v\|_{C(\bar{U})}$. #

<u>Thm:</u> $C(\bar{U})$ is complete, i.e., every Cauchy seq (u_j) in $C(\bar{U})$ converges to some $u \in C(\bar{U})$.

<u>Recall:</u> $u : U \to \mathbb{R}$ is Lipschitz continuous if

$$|u(x) - u(y)| \leq C|x-y| \quad \forall x, y \in U$$

for some constant $C > 0$.

<u>Def:</u> We say that $u : U \to \mathbb{R}$ is Hölder continuous, with Hölder exponent $\gamma \in (0,1]$, if

$$|u(x) - u(y)| \leq C |x-y|^{\gamma} \quad \forall x, y \in U$$

for some constant $C > 0$ ($\gamma = 1$ corresponds to Lipschitz continuity).

Then

$$[u]_{C^{0,\gamma}(\bar{U})} := \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|u(x) - u(y)|}{|x-y|^{\gamma}}$$

is called the Hölder seminorm of u.

<u>Thm:</u> 1). $[\lambda u]_{C^{0,\gamma}(\bar{U})} = |\lambda| [u]_{C^{0,\gamma}(\bar{U})}$,

2). $[u+v]_{C^{0,\gamma}(\bar{U})} \leq [u]_{C^{0,\gamma}(\bar{U})} + [v]_{C^{0,\gamma}(\bar{U})}$.

<u>Pf:</u> 1). $[\lambda u]_{C^{0,\gamma}(\bar{U})} = \sup\limits_{\substack{x,y \in U \\ x \neq y}} \dfrac{|\lambda u(x) - \lambda u(y)|}{|x-y|^{\gamma}} = |\lambda| [u]_{C^{0,\gamma}(\bar{U})}$

2). $$\frac{|(u+v)(x)-(u+v)(y)|}{|x-y|^\gamma} = \frac{|u(x)-u(y)+v(x)-v(y)|}{|x-y|^\gamma}$$

$$\leq \frac{|u(x)-u(y)|}{|x-y|^\gamma} + \frac{|v(x)-v(y)|}{|x-y|^\gamma}$$

$$\leq [u]_{C^{0,\gamma}(\bar{U})} + [v]_{C^{0,\gamma}(\bar{U})},$$

and taking the sup over all $x, y \in U$, $x \neq y$ gives the conclusion. #

$[\cdot]_{C^{0,\gamma}(\bar{U})}$ is not a norm because for any nonzero constant function u, $[u]_{C^{0,\gamma}(\bar{U})} = 0$.

<u>Def:</u> Let $C^{0,\gamma}(\bar{U})$ denote the subspace of $C(\bar{U})$ consisting of functions u for which $[u]_{C^{0,\gamma}(\bar{U})} < \infty$. For $u \in C^{0,\gamma}(\bar{U})$, define the Hölder norm of u by

$$\|u\|_{C^{0,\gamma}(\bar{U})} = \|u\|_{C(\bar{U})} + [u]_{C^{0,\gamma}(\bar{U})}.$$

Thm: $\|\cdot\|_{C^{0,\gamma}(\bar{U})}$ is a norm on $C^{0,\gamma}(\bar{U})$.

Pf: 1). If $\|u\|_{C^{0,\gamma}(\bar{U})} = 0$, then $\|u\|_{C(\bar{U})} = 0$. Since $\|\cdot\|_{C(\bar{U})}$ is a norm, then $u = 0$.

2). For $u \in C^{0,\gamma}(\bar{U})$ and $\lambda \in \mathbb{R}$,

$$\|\lambda u\|_{C^{0,\gamma}(\bar{U})} = \|\lambda u\|_{C(\bar{U})} + [\lambda u]_{C^{0,\gamma}(\bar{U})} = |\lambda| \|u\|_{C(\bar{U})} + |\lambda| [u]_{C^{0,\gamma}(\bar{U})} = |\lambda| \|u\|_{C^{0,\gamma}(\bar{U})}.$$

3). For $u, v \in C^{0,\gamma}(\bar{U})$,

$$\|u+v\|_{C^{0,\gamma}(\bar{U})} = \|u+v\|_{C(\bar{U})} + [u+v]_{C^{0,\gamma}(\bar{U})}$$

$$\leq \|u\|_{C(\bar{U})} + \|v\|_{C(\bar{U})} + [u]_{C^{0,\gamma}(\bar{U})} + [v]_{C^{0,\gamma}(\bar{U})}$$

$$= \|u\|_{C^{0,\gamma}(\bar{U})} + \|v\|_{C^{0,\gamma}(\bar{U})}. \quad \#$$

Thm: $C^{0,\gamma}(\bar{U})$ is complete, i.e., it is a Banach space.

Pf: Let (u_j) be a Cauchy seq in $C^{0,\gamma}(\bar{U})$, i.e.,

$\|u_j - u_k\|_{C^{0,\gamma}(\bar{U})} \to 0$ as $j, k \to \infty$. Since

$$\|u_j - u_k\|_{C^{0,\gamma}(\bar{U})} = \|u_j - u_k\|_{C(\bar{U})} + [u_j - u_k]_{C^{0,\gamma}(\bar{U})}, \quad (*)$$

then $\|u_j - u_k\|_{C(\bar{U})} \to 0$ and $[u_j - u_k]_{C^{0,\gamma}(\bar{U})} \to 0$ as $j, k \to \infty$. So

(u_j) is a Cauchy seq in $C(\bar{U})$. Since $C(\bar{U})$ is complete, then $\exists u \in C(\bar{U})$ st $\|u_j - u\|_{C(\bar{U})} \to 0$. To complete the proof, we will show that $u \in C^{0,\gamma}(\bar{U})$ and $\|u_j \to u\|_{C^{0,\gamma}(\bar{U})} \to 0$.

We have

$$[u]_{C^{0,\gamma}(\bar{U})} = \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|u(x) - u(y)|}{|x-y|^{\gamma}}.$$

Since (u_j) is a Cauchy seq in $C^{0,\gamma}(\bar{U})$, it is bounded in $C^{0,\gamma}(\bar{U})$, i.e.,

$$\|u_j\|_{C^{0,\gamma}(\bar{U})} = \|u_j\|_{C(\bar{U})} + [u_j]_{C^{0,\gamma}(\bar{U})} \leq C \qquad \forall j$$

for some constant $C > 0$. Then

$$[u_j]_{C^{0,\gamma}(\bar{U})} = \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|u_j(x) - u_j(y)|}{|x-y|^\gamma} \leq C.$$

Letting $j \to \infty$ gives $[u]_{C^{0,\gamma}(\bar{U})} \leq C$, so $u \in C^{0,\gamma}(\bar{U})$.

Finally,

$$[u_j - u]_{C^{0,\gamma}(\bar{U})} = \|u_j - u\|_{C(\bar{U})} + [u_j - u]_{C^{0,\gamma}(\bar{U})}$$

and $\|u_j - u\|_{C(\bar{U})} \to 0$, so it only remains to show that

$[u_j - u]_{C^{0,\gamma}(\bar{U})} \to 0$. From $\circledast$, $[u_j - u_k]_{C^{0,\gamma}(\bar{U})} \to 0$ as $j, k \to \infty$, i.e.,

for any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ st

$$j, k \geq N \Rightarrow [u_j - u_k]_{C^{0,\gamma}(\bar{U})} < \varepsilon.$$

We have

$$[u_j - u_k]_{C^{0,\gamma}(\bar{U})} = \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|(u_j - u_k)(x) - (u_j - u_k)(y)|}{|x-y|^\gamma}.$$

Fix $x \neq y$ in U. Then

$$j, k \geq N \Rightarrow \frac{|(u_j - u_k)(x) - (u_j - u_k)(y)|}{|x-y|^\gamma} < \varepsilon.$$

For $j, k \geq N$,

$$\frac{|(u_j - u)(x) - (u_j - u)(y)|}{|x-y|^\gamma} = \frac{|(u_j - u_k)(x) + (u_k - u)(x) - (u_j - u_k)(y) - (u_k - u)(y)|}{|x-y|^\gamma}$$

$$\leq \frac{|(u_j-u_k)(x)-(u_j-u_k)(y)|}{|x-y|^\gamma} + \frac{|(u_k-u)(x)|+|(u_k-u)(y)|}{|x-y|^\gamma}$$

$$< \varepsilon + \frac{2\|u_k-u\|_{C(\bar{U})}}{|x-y|^\gamma}.$$

Letting $k \to \infty$ gives

$$\frac{|(u_j-u)(x)-(u_j-u)(y)|}{|x-y|^\gamma} \leq \varepsilon \quad \forall j \geq N$$

for $x \neq y$ in U. Now taking the sup over all $x \neq y$ gives

$[u_j - u]_{C^{0,\gamma}(\bar{U})} \leq \varepsilon$ for all $j \geq N$. So $[u_j - u]_{C^{0,\gamma}(\bar{U})} \to 0$. #

$C(\bar{U})$, $C^{0,\gamma}(\bar{U})$, $C^1(\bar{U})$, $C^{1,\gamma}(\bar{U})$

$0 < \gamma \leq 1$

Def: Let $C^1(\bar{U})$ denote the linear space of continuously differentiable functions $u: U \to \mathbb{R}$ st

$$\|u\|_{C^1(\bar{U})} := \|u\|_{C(\bar{U})} + \sum_{i=1}^{n} \|u_{x_i}\|_{C(\bar{U})} < \infty.$$

$x = (x_1, \dots, x_n) \in U$

$u_{x_i} = \frac{\partial u}{\partial x_i}$

Thm: $\|\cdot\|_{C^1(\bar{U})}$ is a norm on $C^1(\bar{U})$.

Pf: 1). Clearly, $\|u\| \geq 0$ for all $u \in C^1(\bar{U})$ and $\|0\|_{C^1(\bar{U})} = 0$.

If $\|u\|_{C^1(\bar{U})} = 0$, then $\|u\|_{C(\bar{U})} = 0$ and hence $u = 0$.

2). For $u \in C^1(\bar{U})$ and $\lambda \in \mathbb{R}$,

$$\|\lambda u\|_{C^1(\bar{U})} = |\lambda| \|u\|_{C(\bar{U})} + \sum_{i=1}^{n} |\lambda| \|u_{x_i}\|_{C(\bar{U})} = |\lambda| \|u\|_{C^1(\bar{U})}.$$

3). For $u, v \in C^1(\bar{U})$,

$$\|u+v\|_{C^1(\bar{U})} = \underbrace{\|u+v\|_{C(\bar{U})}}_{\leq \|u\|_{C(\bar{U})} + \|v\|_{C(\bar{U})}} + \sum_{i=1}^{n} \underbrace{\|u_{x_i} + v_{x_i}\|_{C(\bar{U})}}_{\leq \|u_{x_i}\|_{C(\bar{U})} + \|v_{x_i}\|_{C(\bar{U})}}$$

$$\leq \|u\|_{C^1(\bar{U})} + \|v\|_{C^1(\bar{U})}.$$

Thm: $C^1(\bar{U})$ is a Banach space.

Pf: Let (u_j) be a Cauchy seq in $C^1(\bar{U})$, i.e., $\|u_j - u_k\|_{C^1(\bar{U})} \to 0$ as $j, k \to \infty$. Since

$$\|u_j - u_k\|_{C^1(\bar{U})} = \|u_j - u_k\|_{C(\bar{U})} + \sum_{i=1}^{n} \|(u_j)_{x_i} - (u_k)_{x_i}\|_{C(\bar{U})}, \qquad (*)$$

then $\|u_j - u_k\|_{C(\bar{U})} \to 0$ and $\|(u_j)_{x_i} - (u_k)_{x_i}\|_{C(\bar{U})} \to 0$ for $i = 1, \ldots, n$.

So (u_j) is a Cauchy seq in $C(\bar{U})$ and hence $\exists \underline{u} \in C(\bar{U})$

st $\underbrace{\|u_j - u\|_{C(\bar{U})} \to 0}_{\Rightarrow u_j \to u \text{ uniformly in } U}$. Similarly, for $i = 1, \dots, n$, $((u_j)_{x_i})$ is a

Cauchy seq in $C(\bar{U})$ and hence $\exists \underline{g_i} \in C(\bar{U})$ st $\underbrace{\|(u_j)_{x_i} - g_i\|_{C(\bar{U})} \to 0}$.

Since $u_j \to u$ and $(u_j)_{x_i} \to g_i$ uniformly in U, then $\Rightarrow (u_j)_{x_i} \to g_i$ uniformly in U

$u \in C^1(\bar{U})$ and $u_{x_i} = g_i$. Now

$$\|u_j - u\|_{C^1(\bar{U})} = \|u_j - u\|_{C(\bar{U})} + \sum_{i=1}^{n} \|(u_j)_{x_i} - \underbrace{u_{x_i}}_{g_i}\|_{C(\bar{U})} \to 0.$$

#

<u>Def:</u> Let $C^{1,\gamma}(\bar{U})$ denote the space of continuously differentiable functions $u : U \to \mathbb{R}$ st

$$\|u\|_{C^{1,\gamma}(\bar{U})} := \|u\|_{C^1(\bar{U})} + \sum_{i=1}^{n} [u_{x_i}]_{C^{0,\gamma}(\bar{U})} < \infty.$$

Thm: $\|\cdot\|_{C^{1,\gamma}(\bar{U})}$ is a norm on $C^{1,\gamma}(\bar{U})$.

Pf: Same as before. #

Thm: $C^{1,\gamma}(\bar{U})$ is a Banach space.

Pf: Let (u_j) be a Cauchy seq in $C^{1,\gamma}(\bar{U})$, i.e., $\|u_j - u_k\|_{C^{1,\gamma}(\bar{U})} \to 0$ as $j, k \to \infty$. Since

$$\|u_j - u_k\|_{C^{1,\gamma}(\bar{U})} = \|u_j - u_k\|_{C(\bar{U})} + \sum_{i=1}^{n} \|(u_j)_{x_i} - (u_k)_{x_i}\|_{C(\bar{U})}$$

$$+$$

$$+ \sum_{i=1}^{n} \left[(u_j)_{x_i} - (u_k)_{x_i}\right]_{C^{0,\gamma}(\bar{U})}$$

$$= \|(u_j)_{x_i} - (u_k)_{x_i}\|_{C^{0,\gamma}(\bar{U})}$$

$$= \|u_j - u_k\|_{C(\bar{U})} + \sum_{i=1}^{n} \|(u_j)_{x_i} - (u_k)_{x_i}\|_{C^{0,\gamma}(\bar{U})},$$

this implies that (u_j) is a Cauchy seq in $C(\bar{U})$ and $((u_j)_{x_i})$ is a Cauchy seq in $C^{0,\gamma}(\bar{U})$ for $i=1,\dots,n$. Since $C(\bar{U})$ and $C^{0,\gamma}(\bar{U})$ are complete, then $\exists u \in C(\bar{U})$ st $\|u_j - u\|_{C(\bar{U})} \to 0$ and $\exists g_i \in C^{0,\gamma}(\bar{U})$ st $\|(u_j)_{x_i} - g_i\|_{C^{0,\gamma}(\bar{U})} \to 0$.

Then $u_j \to u$ and $(u_j)_{x_i} \to g_i$ uniformly on U, so $u \in C^1(\bar{U})$ and $u_{x_i} = g_i$. $\square$

$$\|u_j - u\|_{C^{1,\gamma}(\bar{U})} = \|u_j - u\|_{C(\bar{U})} + \sum_{i=1}^{n} \|(u_j)_{x_i} - \underbrace{u_{x_i}}_{g_i}\|_{C(\bar{U})}$$

$$+ \sum_{i=1}^{n} [(u_j)_{x_i} - \underbrace{u_{x_i}}_{g_i}]_{C(\bar{U})}$$

$$\longrightarrow 0. \quad \#$$

$$\alpha = (\alpha_1, \dots, \alpha_n), \quad |\alpha| = \alpha_1 + \dots + \alpha_n, \quad \alpha_i \geq 0$$

$$\frac{\partial^\alpha u}{\partial x^\alpha} = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$$

$f_j(x) \longrightarrow f(x)$ uniformly + $f_j'(x) \longrightarrow g(x)$ uniformly

$\Rightarrow f'(x) = g(x)$

$$\underbrace{f_j(x)}_{\to f(x)} = \underbrace{f_j(x_0)}_{\to f(x_0)} + \underbrace{\int_{x_0}^{x} f_j'(y)\,dy}_{\to \int_{x_0}^{x} g(y)\,dy}$$

$$f(x) = f(x_0) + \int_{x_0}^{x} g(y)\,dy$$

$\Rightarrow f'(x) = g(x)$